

*Non-Calculator***Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.****You have 30 minutes.**_____
Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[4\sqrt{2}(\cos 165^\circ + i \sin 165^\circ) \right] \left[3\sqrt{2}(\cos 240^\circ + i \sin 240^\circ) \right]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

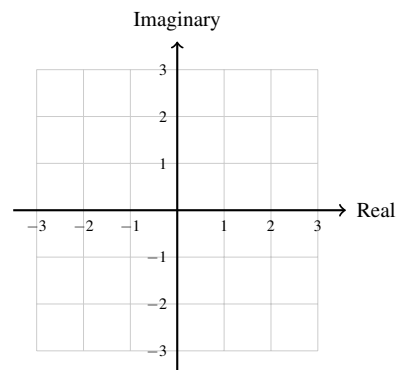
2. $\frac{-3\sqrt{6} + 3i\sqrt{2}}{\sqrt{3} - 3i}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3. $(\sqrt{6} - i\sqrt{2})^5$

Find the fourth roots of the number. Leave answers in trig form. Graph each root as a vector in the complex plane. (3 points)

4. $-8 + 8i\sqrt{3}$



Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

5. $x^3 + 27i = 0$

ANSWER KEY – Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!*Every line shown. Point splits are an estimate of partial credit (halves are possible).*

Find the product. Write the product in rectangular form, using exact values. (3 points)

Problem: $\left[4\sqrt{2}(\cos 165^\circ + i \sin 165^\circ)\right] \left[3\sqrt{2}(\cos 240^\circ + i \sin 240^\circ)\right]$

1. Multiply the r 's, add the θ 's.

$$\begin{aligned} r &= (4\sqrt{2})(3\sqrt{2}) = 12 \cdot 2 = 24 & \theta &= 165^\circ + 240^\circ = 405^\circ \\ 405^\circ - 360^\circ &= 45^\circ \quad (\text{keep the angle between } 0^\circ \text{ and } 360^\circ) \\ 24(\cos 45^\circ + i \sin 45^\circ) &= 24\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) \\ &= \boxed{12\sqrt{2} + 12\sqrt{2}i} \end{aligned}$$

*Trap: $\sqrt{2} \cdot \sqrt{2} = 2$, so $r = 12 \cdot 2 = 24$. Leaving $12\sqrt{4}$, or writing $12\sqrt{2}$, costs the point.**1 pt r , 1 pt reduced angle, 1 pt exact rectangular.*

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

Problem: $\frac{-3\sqrt{6} + 3i\sqrt{2}}{\sqrt{3} - 3i}$

2. Convert the top and the bottom separately (sketch each for the quadrant).

Top: $r = \sqrt{(-3\sqrt{6})^2 + (3\sqrt{2})^2} = \sqrt{54 + 18} = \sqrt{72} = 6\sqrt{2}$ $\tan \theta = \frac{3\sqrt{2}}{-3\sqrt{6}} = -\frac{1}{\sqrt{3}} \Rightarrow \text{ref } 30^\circ, \text{ QII} \Rightarrow 150^\circ$

Bottom: $r = \sqrt{(\sqrt{3})^2 + (-3)^2} = \sqrt{3 + 9} = \sqrt{12} = 2\sqrt{3}$ $\tan \theta = \frac{-3}{\sqrt{3}} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIV} \Rightarrow 300^\circ$

$$\begin{aligned} \frac{6\sqrt{2} \text{ cis } 150^\circ}{2\sqrt{3} \text{ cis } 300^\circ} &= \frac{3\sqrt{2}}{\sqrt{3}} \text{ cis}(150^\circ - 300^\circ) = \frac{3\sqrt{6}}{3} \text{ cis}(-150^\circ) = \sqrt{6} \text{ cis}(-150^\circ + 360^\circ) = \sqrt{6} \text{ cis } 210^\circ \\ &= \sqrt{6} \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = -\frac{\sqrt{18}}{2} - \frac{\sqrt{6}}{2}i = \boxed{-\frac{3\sqrt{2}}{2} - \frac{\sqrt{6}}{2}i} \end{aligned}$$

Trap: $(-3\sqrt{6})^2 = 9 \cdot 6 = 54$ (square the 3 too). Negative angle: add 360° . Rationalize $\frac{3\sqrt{2}}{\sqrt{3}}$ and simplify $\sqrt{18} = 3\sqrt{2}$.

Find the power. Write the answer in rectangular form, using exact values. (3 points)

Problem: $(\sqrt{6} - i\sqrt{2})^5$

3. Convert, then De Moivre: $[r \operatorname{cis} \theta]^n = r^n \operatorname{cis}(n\theta)$.

$$r = \sqrt{(\sqrt{6})^2 + (-\sqrt{2})^2} = \sqrt{6+2} = \sqrt{8} = 2\sqrt{2}$$

$$\tan \theta = \frac{-\sqrt{2}}{\sqrt{6}} = -\frac{1}{\sqrt{3}} \Rightarrow \text{ref } 30^\circ, \text{ QIV} \Rightarrow 330^\circ$$

$$(2\sqrt{2} \operatorname{cis} 330^\circ)^5 = (2\sqrt{2})^5 \operatorname{cis}(5 \cdot 330^\circ) = 2^5 (\sqrt{2})^5 \operatorname{cis} 1650^\circ$$

$$= 32 \cdot 4\sqrt{2} \operatorname{cis} 1650^\circ = 128\sqrt{2} \operatorname{cis} 1650^\circ$$

$$1650^\circ - 4(360^\circ) = 1650^\circ - 1440^\circ = 210^\circ$$

$$128\sqrt{2}(\cos 210^\circ + i \sin 210^\circ) = 128\sqrt{2} \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = \boxed{-64\sqrt{6} - 64\sqrt{2}i}$$

Trap: $(\sqrt{2})^5 = \sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2} = 4\sqrt{2}$, not $5\sqrt{2}$ or $2\sqrt{5}$.

Find the fourth roots of the number. Leave answers in trig form. Graph each root as a vector in the complex plane. (3 points)

Problem: $-8 + 8i\sqrt{3}$

4. Convert, then the n th root theorem: $\sqrt[n]{r} \operatorname{cis} \left(\frac{\theta}{n} + \frac{360^\circ k}{n} \right), k = 0, 1, \dots, n-1$.

$$r = \sqrt{(-8)^2 + (8\sqrt{3})^2} = \sqrt{64 + 192} = \sqrt{256} = 16$$

$$\tan \theta = \frac{8\sqrt{3}}{-8} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QII} \Rightarrow 120^\circ$$

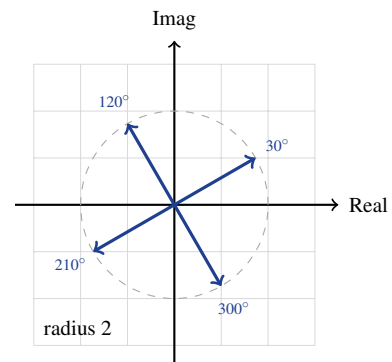
$$\sqrt[4]{16} = 2 \quad \text{first angle } \frac{120^\circ}{4} = 30^\circ$$

$$\text{spacing } \frac{360^\circ}{4} = 90^\circ$$

$$\boxed{2 \operatorname{cis} 30^\circ, 2 \operatorname{cis} 120^\circ, 2 \operatorname{cis} 210^\circ, 2 \operatorname{cis} 300^\circ}$$

Stop at four roots: the next one, 390° , is 30° again.

Trap: $\sqrt{256} = 16$, then $\sqrt[4]{16} = 2$ (not 4).



four equal vectors, 90° apart

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

Problem: $x^3 + 27i = 0$

5. Isolate x^3 first, then take cube roots of the constant.

$$x^3 + 27i = 0 \Rightarrow x^3 = -27i = 0 - 27i \quad (\text{negative imaginary axis})$$

$$r = 27, \quad \theta = 270^\circ \quad x = (27 \operatorname{cis} 270^\circ)^{1/3}$$

$$\sqrt[3]{27} = 3 \quad \frac{270^\circ}{3} = 90^\circ \quad \text{spacing } \frac{360^\circ}{3} = 120^\circ$$

$$x = 3 \operatorname{cis} 90^\circ, 3 \operatorname{cis} 210^\circ, 3 \operatorname{cis} 330^\circ$$

$$3 \operatorname{cis} 90^\circ = 3(0 + i) = 3i$$

$$3 \operatorname{cis} 210^\circ = 3\left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = -\frac{3\sqrt{3}}{2} - \frac{3}{2}i$$

$$3 \operatorname{cis} 330^\circ = 3\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right) = \frac{3\sqrt{3}}{2} - \frac{3}{2}i$$

$$x = 3i, -\frac{3\sqrt{3}}{2} - \frac{3}{2}i, \frac{3\sqrt{3}}{2} - \frac{3}{2}i$$

Check one: $(3i)^3 = 27i^3 = 27(-i) = -27i \checkmark$

Trap: moving $+27i$ across makes it $-27i$, which is 270° , not 90° . Using 90° gives the wrong three roots.